

MidTerm Presentation 2 :

Approximate location a compromise between privacy and utility

December 11, 2025 - EP2510 Advanced Network System Security

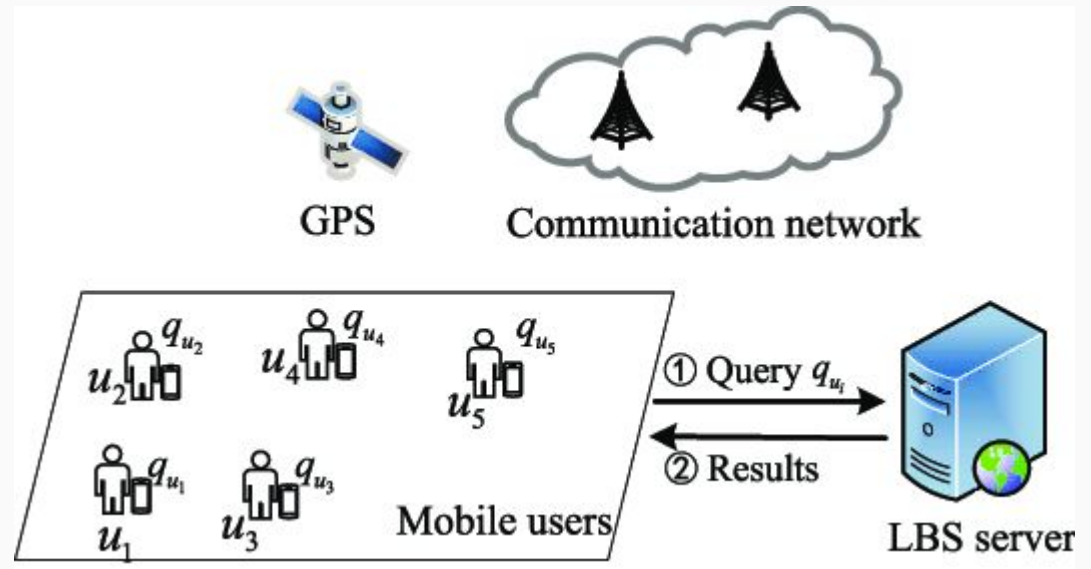
Plan

I] Actors and possible threats

II] Related Work

III] My future work

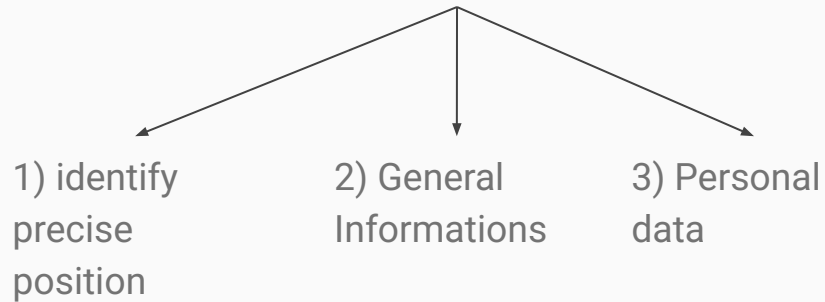
System



- GPS System
- Communication System
- LBS

Attack Model

Server compromised => Honest-but-curious server



Location Privacy Matters

A lot of services

Smart Home : Nest Thermostat / Google Home

Navigation : Google maps

Fitness : strava, Nike run

Transportation : Uber / Bolt /Lime

Weather ; Dating

Know a lot of things

Preferences (temperatures)

Habits (where you live, Favorite bakery, etc)

[location of secret US army bases](#)

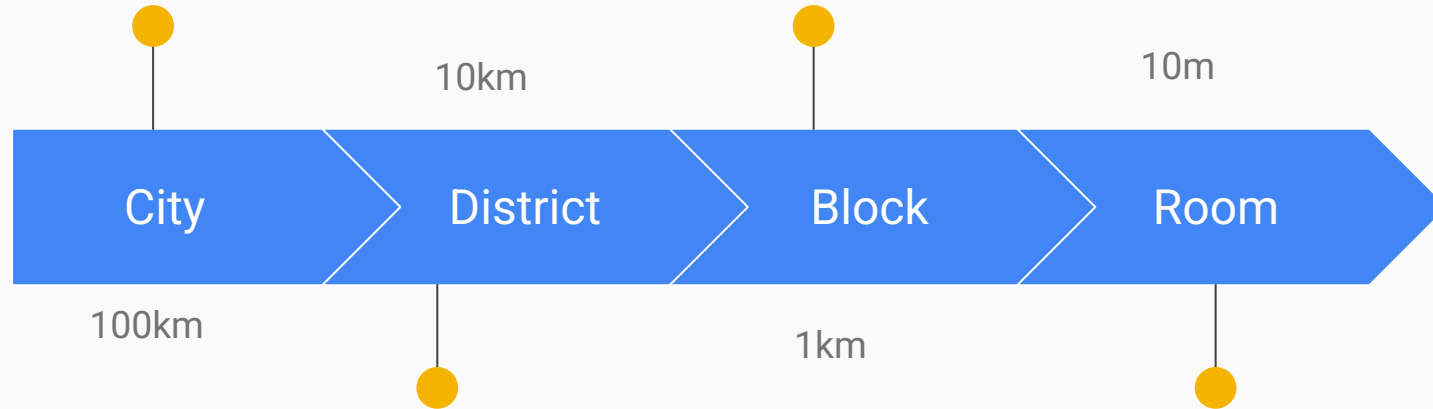
Location (car theft, house burglary)

Car theft

House burglary

Create heat map of
city

Personal data



Advertising

Impersonation

Victim habits

Related Works

<https://www.sciencedirect.com/topics/computer-science/location-privacy>

Liyue Fan and Ishan Gote. 2021. A Closer Look: Evaluating Location Privacy Empirically. In Proceedings of the 29th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '21). Association for Computing Machinery, New York, NY, USA, 488-499. <https://doi.org/10.1145/3474717.3484219>

Sven Heitmann, Alexia Pagotto & Christian Kray (2025) Approximate vs. precise location in popular location-based services, Journal of Location Based Services, 19:1, 1-42, DOI: [10.1080/17489725.2024.2310006](https://doi.org/10.1080/17489725.2024.2310006)

Related Work 1

Sven Heitmann, Alexia Pagotto & Christian
Kray (2025) Approximate vs. precise
location in popular location-based
services, Journal of Location Based
Services, 19:1, 1-42, DOI:
10.1080/17489725.2024.2310006

(paper seen during Midterm
presentation)

Which LBS is used ?

36 iPhones + 36 Android => evaluate around 30 apps (Bolt, Doctolib, Uber ...)

Conduct of study

Allow fine location on app to understand how it is supposed to work; switch to approximate location to view changes.

Calculus are subjective but creation of soft goals and hard goals

Utility vs privacy

All in Germany ; difference between iOS & Android ;

doesn't do a correlation between privacy and coarse location

Related Work 2

Liyue Fan and Ishan Gote. 2021. A Closer Look: Evaluating Location Privacy Empirically. In Proceedings of the 29th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '21). Association for Computing Machinery, New York, NY, USA, 488–499.
<https://doi.org/10.1145/3474717.348421>

9

Which LBS is used ?

RioBuses and geoLife for Data (from poinA to point B)

Conduct of study

Developed a java program that process data and create metrics, especially “utility” (which is a set of number) for SpotMe,Cloaking,Noise techniques

Utility vs privacy

Utility is a set of measure

Privacy not well-defined -> if there is noise... there is already some privacy

Utility/Params
Hamming
Haversine (in m)
Tot Dist (in %)
Max Dist (in %)
Std Dev Displacement (in %)
Max Dist Home (in %)
Rad Gyration (in %)
Diff Places (in %)
Significant Places (in %)
Avg Mobility Error (in %)

Work with trips, “continuous” points,

Hypothesis

The user allows the LBS to know where he wants to go

BUT not where he is.

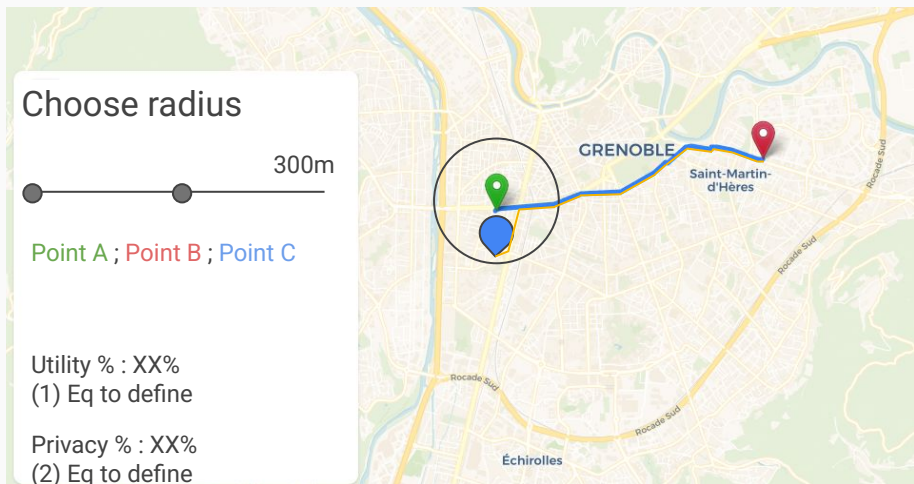
He knows almost where he is on a map and knows what is the main direction to go.

BUT he needs to have a path to follow.

We want to find if there are specific configuration where approximate location doesn't impact the service (utility % > 90% for exemple)

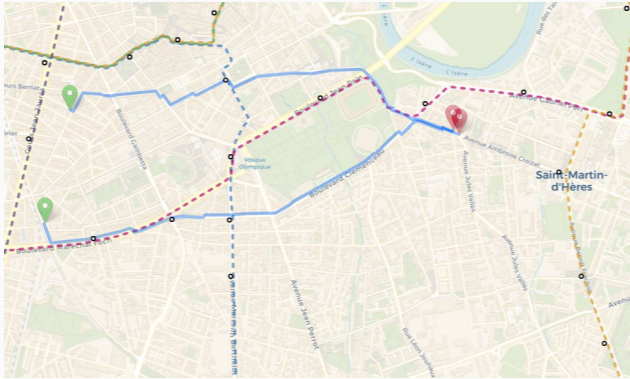
What I aim to work on

- Find equations
 - Privacy formula with 2 points
 - Utility formula with 3 points
- See how to implement with openStreetMap
- Make the possibility to change the radius of approximate location to know how it changes



Ideas for formulas

Utility % : %percentage of shared path



Privacy % : $1 - \exp(-\text{radius} * d(A, C))$

Where d is the haversine distance

Exponential because the further the more private you are